

# CBD Definition for EMS Optimization Study

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## Overview

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The Central Business District (CBD) for this study is defined based on the **MTA Congestion Relief Zone** boundary, which closely aligns with the economic core of Manhattan below 60th Street. This boundary was chosen because:

1. **Official designation** – The MTA congestion pricing zone is a legally defined area with a precise geographic boundary
2. **High activity density** – It encompasses the highest density of commercial, retail, and transit activity in Manhattan
3. **EMS relevance** – The CBD experiences significantly higher crash rates due to traffic density, pedestrian volume, and complex intersection geometry

## CBD Boundary

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The geographic boundary is stored in `data/raw/cbd_boundary.geojson` as a MultiPolygon geometry in WGS84 (EPSG:4326).

### Approximate extent:

- **North boundary:** ~60th Street
- **South boundary:** Battery Park / Financial District
- **East boundary:** East River waterfront
- **West boundary:** Hudson River waterfront (with exclusions for some waterfront areas)

## CBD Precincts

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Police precincts were classified as CBD if  $\geq 30\%$  of their area overlaps the CBD boundary. This threshold ensures that precincts with meaningful CBD presence are included while excluding those with only marginal overlap.

| Precinct | CBD Overlap | Description                         |
|----------|-------------|-------------------------------------|
| 1        | 100.0%      | Financial District / Tribeca        |
| 5        | 100.0%      | Chinatown / Lower East Side (south) |
| 6        | 100.0%      | West Village / Greenwich Village    |
| 7        | 100.0%      | Lower East Side                     |
| 9        | 100.0%      | East Village                        |
| 10       | 100.0%      | Chelsea (south)                     |
| 13       | 100.0%      | Gramercy / Stuyvesant Town          |
| 14       | 100.0%      | Midtown South                       |
| 17       | 100.0%      | Midtown East                        |
| 18       | 61.4%       | Midtown North (partial)             |

**Total CBD precincts:** 10 out of 25 Manhattan precincts (40%)

## Demand Characteristics

Based on analysis of 416,434 Manhattan crash records:

- **CBD crashes:** 231,786 (55.7% of all Manhattan crashes)
- **Non-CBD crashes:** 184,648 (44.3%)
- **CBD demand intensity:** ~1.26× higher per-precinct than non-CBD average

The CBD generates disproportionately high demand relative to its geographic area, making it a critical focus for EMS resource allocation.

## Methodology

The CBD precinct classification was performed using:

1. Loading the MTA Congestion Relief Zone boundary ( `data/raw/cbd_boundary.geojson` )
2. Loading police precinct boundaries ( `data/raw/Police_Precincts_20260223.csv` )
3. Computing the intersection area ratio for each Manhattan precinct
4. Classifying precincts with  $\geq 30\%$  overlap as CBD

This classification is used in:

- CBD robustness experiments ( `scripts/run_cbd_experiment.py` )
- CBD analysis ( `notebooks/09_cbd_analysis.ipynb` )
- Technical report CBD robustness section